

MAK CS

Translators

Translations

- Software's used to convert Source code ^{→ Actual Programming code.} into object code.
 - Machine Code
 - Ready for execution
 - Understand by CPU
 - Binary Form

1. Compiler

2. Interpreter

3. Java Virtual Machines

Compiler

- Source Code is provided to compiler.
 - Initially no data is provided.
- Code is analyzed.
- First Line of code is READ, converted into Intermediate Code.
 - Error Free.
- IF a Line having an Error, Error is recorded and we move to the next line until whole code is completed.
- At end of code, Contin Errors, they are reported — Error Report Generated.
- If code having no errors, an Executable File is generated. Whole code is converted in one go.

Advantages

- Always generate an executable file
- Executable files cannot be altered since no source code is available.
- Mostly they are Error Free.
- Only Executable file shared, easy to distribute — No source code.
- Executable file can run on different machines
- Conversion is Fast, since whole code in go.

Disadvantages

- If .exe file is not tested properly, can crash memory.
- Since the whole code is converted in one go, memory can CRASH.
- Finding Errors becomes complex

Interpreters

- Source Code along data is provided
- Code is analyzed.
- First Line of code is Read and Executed, if no Errors.
- IF a Line contain Error, Execution is halted
- Some steps are performed until whole code is converted.

Advantages

- Finding and Removing Errors is Easy
- Cannot crash memory.
- If source code is purchased, interpreter comes along with code.

Disadvantages

- It takes Longer time for execution
- To share code, source code is provided, alterations can be done.
- To execute code, always need interpreter.
- Code running on one machine cannot be interpreted on another.

JAVA VIRTUAL MACHINES

- Programs are partially compiled (**BYTECODE**) and partially interpreted
- Hybrid approach and offers portability due to byte code format.
- Platform independent
- Partially compiled programs can be used on different platforms as they are interpreted when RUN
- Code is optimised as machine code is generated at RunTime.
- Source code does not need to be recompiled, more efficient to RUN.

IDE --- Integrated Development Environment

- Softwares in which we perform coding.
- Programs are written and tested by IDEs
- Source Code Editor
- Compiler and Interpreter
- Debugging
- Auto documentation.

Pretty Print

- Displaying or printing well set of formatted source code.
- Easier to read and understand.
- Text columns are changed, underline as well.

Debugging :

- Finding Logical Errors by Running the code and Tracking the variables in variable window watch.

Break Points

- A deliberate pause in the execution of a program during testing so that we can observe values of variables.

Auto documentor

- Provide information about different function
- How to implement certain functions.

Report Window / Watch Window

- Separate window, in Runtime that shows the contents of variables updating during execution of program.

WRITE CODE

- Editor
- Pretty Print
- Expand and collapse into different blocks
- Dynamic syntax check
- Auto formatting
- Context sensitive prompts, help to complete statements

Testing Code

- Stepping — Single
- Break points
- Debugger
- Report window